THE MAGNITUDE OF CATEGORIES OF TEXTS ENRICHED BY LANGUAGE MODELS

TAI-DANAE BRADLEY AND JUAN PABLO VIGNEAUX

Abstract. The purpose of this article is twofold. Firstly, we use the next-token probabilities given by a language model to explicitly define a $[0,1]$-enrichment of a category of texts in natural language, in the sense of Bradley, Terilla, and Vlassopoulos. We consider explicitly the terminating conditions for text generation and determine when the enrichment itself can be interpreted as a probability over texts. Secondly, we compute the Möbius function and the magnitude of an associated generalized metric space $\mathcal{M}$ of texts using a combinatorial version of these quantities recently introduced by Vigneaux. The magnitude function $f(t)$ of $\mathcal{M}$ is a sum over texts x (prompts) of the Tsallis t-entropies of the next-token probability distributions $p(-|x)$ plus the cardinality of the model's possible outputs. The derivative of f at $t = 1$ recovers a sum of Shannon entropies, which justifies seeing magnitude as a partition function. Following Leinster and Schulman, we also express the magnitude function of $\mathcal{M}$ as an Euler characteristic of magnitude homology and provide an explicit description of the zeroeth and first magnitude homology groups.

1. Introduction

In recent years, advances in large language models (LLMs) have brought renewed attention to mathematical structures underlying language. Building on the observation that LLMs acquire a great deal of semantic information through the contexts and statistical co-occurrences of fragments of text, the first author together with Terilla and Vlassopoulos described a category-theoretical framework for mathematical structure inherent in text corpora in [?]. This involves a category of strings from a finite alphabet of symbols, with morphisms indicating substring containment; the strings correspond to texts or fragments of texts in a language. Statistical information is incorporated through an enrichment of this category over the unit interval by assigning a value $\pi(y|x) \in [0,1]$ to each pair of strings x and y, which can intuitively be thought of as the probability that y is an extension of an input x. By further embedding this enriched category of strings into an enriched category of copresheaves valued in the unit interval, a setting which contains rich mathematical structure, one can further explore semantic information via enriched analogues of categorical limits and colimits, which are akin to logical operations on meaning representations of texts. Moreover, as shown in both [?] and the more recent [?], one can alternatively enrich over the extended non-negative reals $[0,\infty]$ to obtain a generalized metric space yielding additional geometric insights.

While an explicit definition of $\pi(y|x)$ was not given in [?], we will show in this article that these values may indeed arise from next-token probabilities generated

Date: March 14, 2025.

by an LLM. Although a similar construction appears in [**?**], ours takes into account finite context size as well as beginning- and end-of-sentence tokens (denoted respectively $\perp$ and $\dagger$). This allows us to prove that $\pi(-|x)$ can be regarded as a probability mass function on the set $T(x)$ of *terminating states* of an LLM with input x or, equivalently, over the possible outputs. As we discuss below, the set $T(x)$ consists of extensions of x that end with $\dagger$ or reach the maximum size.

Moreover, we extend the geometric perspective by computing a fundamental invariant of enriched categories—their *magnitude*. Magnitude generalizes the cardinality of a set and the Euler characteristic of a topological space, and it can be regarded as the "size" of an enriched category [**?**, **?**]. Lawvere [**?**] observed that one can see a $([0,\infty], \geq, +, 0)$-enriched category as a *generalized metric space*, with a distance function that may be nonsymmetric and degenerate but still satisfies the triangle inequality. The computation of magnitude of ordinary metric spaces under this identification reveals rich connections with traditional invariants from integral geometry and geometric measure theory such as volume, capacity, dimension, and intrinsic volumes [**?**].

In this article, we study the generalized metric space $\mathcal{M}$ whose elements are strings $x, y, \ldots$ and whose pairwise distances are given by $d(x,y) = -\ln \pi(y|x)$. We use a novel combinatorial formula introduced by Vigneaux in the recent article [**?**] to compute the Möbius coefficients of $\mathcal{M}$, see Proposition **??**. Then we use these Möbius coefficients to calculate the *magnitude function* $f(t) := \mathrm{Mag}(t\mathcal{M})$, for $t > 0$. According to Proposition **??**,

$$f(t) = (t-1) \sum_{x \in \mathrm{ob}(\mathcal{M}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)),$$

where

(1) $\#(T(\perp))$ is the cardinality of the set of terminating states for the prompt $\perp$,
(2) p_x is the probability distribution over tokens generated by a model prompted by x, and
(3) H_t is the Tsallis t-entropy: $H_t(p_1, \ldots, p_m) = (1 - \sum_{i=1}^m p_i^t)/(t-1)$.

The derivative of f at $t = 1$ is a sum of Shannon entropies: $\sum_{x \in \mathrm{ob}(\mathcal{M}) \setminus T(\perp)} H(p_x)$. Analogous results holds for the subspace $\mathcal{M}_x$ of texts that extend a given prompt x, see Remark **??**.

The dependence of magnitude on the terminating states is reminiscent of similar results pertaining to posets or metric spaces that show a dependency on the "boundary." For instance, if a finite poset has top and bottom elements, then its magnitude can be computed by evaluating its Möbius function (defined in Section **??**) at those extremal elements [**?**], [**?**, Proposition 3.8.5]. And in the context of metric spaces, *weighting vectors* [**?**]—such as $(\sum_y \mu(x,y))_x$ when the Möbius coefficients μ exist—effectively detect the boundary of certain subsets of Euclidean space, an insight with recent applications in machine learning [**?**, **?**]; see also [**?**].

Finally, we show in Proposition **??** that the magnitude function can be expressed in terms of the ranks of the magnitude homology groups

$$f(t) = \sum_{\ell} e^{-t\ell} \sum_{k \geq 0} (-1)^k \,\mathrm{rank}(H_{k,\ell}(\mathcal{M})),$$

where the first sum ranges over those ℓ that may appear as finite lengths of paths in $\mathcal{M}$. This result mirrors [?, Theorem 7.14], although our proof follows directly from Proposition ?? and encounters significantly less technical complications.

Structure of the article. Section ?? derives, from a given LLM, an enriched category of strings and an associated generalized metric space. Subsection ?? introduces the basic definitions concerning tokens and texts, including the partial order of texts. Subsection ?? describes the process of text generation by an LLM, defines the function π, and shows that $\pi(-|x)$ is a probability mass function over $T(x)$. Subsection ?? reminds the reader of some basic definitions from enriched category theory, focusing on categories enriched over commutative monoidal preorders, which are a particularly simple case. Subsection ?? shows that π defines a $[0,1]$-category $\mathcal{L}$ of texts in the sense of [?]. Subsection ?? introduces the corresponding generalized metric space $\mathcal{M}$ using the isomorphism of categories $-\ln : [0,1] \to [0,\infty]$.

Section ?? is dedicated to magnitude. Subsection ?? gives a short overview of categorical magnitude and Subsection ?? treats the classical case of posets. Subsection ?? contains some basic definitions about digraphs. These are used in Subsection ?? to compute the Möbius coefficients and magnitude of $\mathcal{M}$. Subsection ?? covers the magnitude homology of $\mathcal{M}$. Finally, Section ?? presents some final remarks and perspectives.

2. Obtaining Enriched Categories from an LLM

2.1. **Tokens and texts.** Let A be a finite alphabet (e.g. a set of tokens) and consider all finite strings $a, b, \dots$ from the free monoid A^*. In LLMs, tokens correspond to words, pieces of words, or special characters, hence longer texts (e.g. sentences, paragraphs) are elements of A^*. Below, we call elements of A *tokens* and elements of A^* *strings* or *texts*. Write $a \leq b$ whenever a is a *prefix* of b, that is, whenever $b = aa'$ for some $a' \in A^*$. This defines a partial order since $\leq$ is reflexive (as A^* contains the empty string ϵ), transitive, and antisymmetric. We denote by $|x|$ the length of a string $x \in A^*$.

Besides the elements of A, we introduce two special tokens $\perp$ and $\dagger$ which represent, respectively, the so-called “beginning-of-sentence” and “end-of-sentence” tokens. Unlike the elements A, the tokens $\perp$ and $\dagger$ do not appear in arbitrary positions; their role is clarified below. Although some systems conflate both symbols, it is useful to keep the distinction here.

We regard a partially ordered set $(P, \leq)$ as a thin category with object set P and such that $x \to y$ if and only if $x \leq y$. Given $N \in \mathbb{N}$, we define $\mathsf{L} := \mathsf{L}^{\leq N}$ as the full subcategory of $((A \cup \{\perp, \dagger\})^*, \leq)$ made of strings that start with $\perp$ and are followed by at most $N-1$ symbols, all of them in A with the exception of the last one, which might also equal $\dagger$; in other terms,

$$\mathrm{ob}(\mathsf{L}) = \{\perp a : a \in A^* \text{ and } |a| \leq N-1\} \sqcup \{\perp a \dagger : a \in A^* \text{ and } |a| < N-1\}.$$

More explicitly, given objects x, y of L, there is a morphism $x \to y$ whenever $x \leq y$ as elements of $(A \cup \{\perp, \dagger\})^*$, that is, if x is a prefix of y. We refer to an object of the form $\perp a$ as an *unfinished text* and to an object of the form $\perp a \dagger$ as a *finished text*. For any non-identity morphism $x \to y$, the prefix x is necessarily an unfinished text, whereas y can be finished or unfinished. Observe that L has an initial object,[1]

[1]By this point, the reader may have wondered why $\dagger$ is used as the end-of-sentence symbol instead of $\top$, given our use of $\perp$ for the beginning-of-sentence symbol. To start, the category

which is $\perp$, and also that it has a natural "grading," in the sense that it has a height function given by the length $|x|$ of strings x.

It will be convenient to write the set $\mathrm{ob}(\mathsf{L})$ as a disjoint union $\bigsqcup_{i=0}^{N-1} \mathsf{L}^{(i)}$, where $\mathsf{L}^{(j)}$ consists of strings of length $1+j$. Aside from identities, the arrows of L only go from elements of $\mathsf{L}^{(i)}$ to elements of $\mathsf{L}^{(j)}$ for $j > i$. Similarly, given a string x in L, there is a full subcategory L_x of L whose objects are $y \in \mathsf{L}$ such that $x \to y$; in this case, we have $\mathrm{ob}(\mathsf{L}_x) = \bigsqcup_{i=0}^{N-|x|} \mathsf{L}_x^{(j)}$, where $\mathsf{L}_x^{(j)}$ consists only of those strings of length $|x|+j$, namely, those strings that extend x on the right by j tokens. So, $\mathsf{L}_x^{(0)} = \{x\}$ and $\mathsf{L}_x^{(1)} = \{xa_1 : a_1 \in A \cup \{\dagger\}\}$ and $\mathsf{L}_x^{(2)} = \{xa_1a_2 : a_1 \in A, a_2 \in A \cup \{\dagger\}\}$ and so on. Remark that $\mathsf{L}_\perp^{(j)} = \mathsf{L}^{(j)}$ for any $j \geq 0$. And if x is a finished text, then L_x has a single object, namely x.

We will refer to L as a poset or as a category interchangeably, since the two notions coincide here. Either way, it keeps track of which strings are right extensions of other strings in a language. To incorporate statistical information, we turn to enriched category theory in the following sections.

2.2. **LLMs and induced probabilities on texts.** We characterize the behavior of a probabilistic large language model trained on next-token prediction (e.g. GPT, Llama) as follows. For any user input $a \in A^*$, the model is fed the tokenized prompt $x = \perp a$ and generates a probability distribution $p_x := p(-|x) \colon A \cup \{\dagger\} \to [0,1]$. After generating p_x, the model then samples a token a_1 according to p_x. If $a_1 = \dagger$, then the process terminates and the model outputs a. Otherwise, the token a_1 is appended to a. These steps are then repeated. The model generates a probability distribution $p_{\perp aa_1}$ on $A \cup \{\dagger\}$, samples a token a_2 according to it, and either terminates execution if $a_2 = \dagger$ (outputting aa_1) or appends a_2 to aa_1 to generate $p_{\perp aa_1a_2}$, and so on.

We assume that the process stops when $\dagger$ is sampled or when the extended prompt $y = xa_1 \cdots a_{N-|x|}$ reaches a maximum length N; we call N the *cutoff.* We can think of $N-1$ as the context size of the language model, i.e. the maximum length of a string x that can be used to generate a distribution p_x. Consequently, we define the set of *terminating states* for the input $x = \perp a$ as follows.

Definition 1. The set of *terminating states* of an unfinished text x in L is defined to be

$$T(x) = \{y \in \mathrm{ob}(\mathsf{L}_x) : y \text{ is an unfinished text of length } N \text{ or } y \text{ is a finished text such that } |y| \leq N\}. \tag{1}$$

In the first case, $y = xa'$ for some $a' \in A^*$ such that $|a'| = N - |x|$; for such a terminating state the model outputs aa' to the user. In the second case, $y = xa''\dagger$ for some $a'' \in A^*$ such that $|a''| \leq N - |x| - 1$; for such a terminating state the model outputs aa''. Either way, there is a bijective correspondence between the outputs of the model and the terminating states.[2]

L does not have a terminal object, so we wish to avoid using notation that suggests otherwise. Further, one might think of the end-of-sentence token as the state which "kills" the generation of a string, hence a dagger.

[2]Notice that this description is only valid for prompts whose length is strictly less than N: to extend a prompt x of length greater than $N-1$, one needs to produce first a subrogate prompt $f(x)$ such that $|f(x)| \leq N-1$ and feed it to the language model to generate $p_{f(x)}$ and sample a token.

Now, it is tempting to define the probability of the model's output b given the user input a as the product of the intermediate probabilities of the tokens involved in its production. For example, when $N > 5$, the probability of generating the output $b = a_1a_2a_3a_4a_5 \in A^*$ given the user's prompt $a = a_1a_2$ would be

$$p(a_3|\perp a)p(a_4|\perp aa_3)p(a_5|\perp aa_3a_4)p(\dagger|\perp aa_3a_4a_5).$$

However, it is not obvious that this rule indeed defines a probability mass function over some set. We prove this below. But first, let us define these products in general.

Definition 2. For any objects x and y of L, define

$$\pi(y|x) := \begin{cases} 1 & \text{if } x = y \\ 0 & \text{if } x \not\to y \\ \displaystyle\prod_{i=1}^{k} p(a_{t+i}|y_{<t+i}) & \text{if } x \to y \end{cases}. \tag{2}$$

In the third case, we assume that $x = \perp a_1 \cdots a_t$ and $y = xa_{t+1} \cdots a_{t+k}$ for some $k \geq 1$, $(a_i)_{i=1}^{t+k-1} \subset A$, and $a_{t+k} \in A \cup \{\dagger\}$. The symbol $y_{<t+i}$ denotes the string $\perp a_1 \cdots a_{t+i-1}$, and $y_{<t+1} = x$.

For a given string x in L, the function $\pi(-|x)$ is not a probability mass function on its whole support. (To begin with, $\pi(x|x) = 1$.) Nonetheless, it becomes a probability mass function when restricted to $T(x)$.

Proposition 1. *Given an unfinished text x in L, the function $\pi(-|x)|_{T(x)}$ is a probability mass function on the terminating states of the input x.*

Proof. We prove the claim by induction on $m = N - |x|$, the largest number of symbols that can succeed x until reaching the model's cutoff. If $m = 0$, then $T(x) = \{x\}$ and by definition $\pi(x|x) = 1$. If $m = 1$, then $T(x) = \{xa \mid a \in A \cup \{\dagger\}\}$ and

$$\sum_{y \in T(x)} \pi(y|x) = \sum_{a \in A \cup \{\dagger\}} p_x(a) = 1,$$

since p_x is assumed to be a probability mass function on $A \cup \{\dagger\}$. For general $m \geq 1$,

$$\sum_{y \in T(x)} \pi(y|x) = \sum_{i=0}^{m-1} \sum_{a \in A^i} \pi(xa\dagger|x) + \sum_{a \in A^m} \pi(xa|x) \tag{3}$$

$$= \sum_{i=0}^{m-1} \sum_{a \in A^i} \pi(xa\dagger|x) + \sum_{\substack{a = a'a'' \\ a' \in A^{m-1}, a'' \in A}} p(a''|xa')\pi(xa'|x) \tag{4}$$

$$= \sum_{i=0}^{m-2} \sum_{a \in A^i} \pi(xa\dagger|x) + \sum_{a' \in A^{m-1}} \pi(xa'|x) \sum_{a'' \in A \cup \{\dagger\}} p(a''|xa') \tag{5}$$

$$= \sum_{i=0}^{m-2} \sum_{a \in A^i} \pi(xa\dagger|x) + \sum_{a' \in A^{m-1}} \pi(xa'|x) \tag{6}$$

The two sums in (**??**) follow from the definition of $T(x)$ and π, where the first sum accounts for finished texts (i.e. those ending in $\dagger$), and the second sum accounts for unfinished texts of length N. We obtain (**??**) by identifying $a \in A^m$ with $(a', a'') \in A^{m-1} \times A$. Then (**??**) follows by rewriting $\pi(xa\dagger|x)$ as $\pi(xa|x)p(\dagger|xa)$ and reordering the terms. The expression in (**??**) equals one in virtue of the induction hypothesis. □

2.3. **Preliminaries on enriched category theory.** We shall use π to derive an enriched category of strings $\mathcal{L}$ from L. For the convenience of the reader, we give here the definition of a category enriched over a commutative monoidal preorder [**?**, Chapter 2], which is the only case we will need. See also [**?**].

Definition 3. A *commutative monoidal preorder* $(\mathcal{V}, \leq, \otimes, 1)$ is a preordered set $(\mathcal{V}, \leq)$ and a commutative monoid $(\mathcal{V}, \otimes, 1)$ satisfying $x \otimes y \leq x' \otimes y'$ whenever $x \leq x'$ and $y \leq y'$.

Definition 4. Let $(\mathcal{V}, \leq, \otimes, 1)$ be a commutative monoidal preorder. A (small) *category enriched over* $\mathcal{V}$, or simply a $\mathcal{V}$*-category*, $\mathcal{C}$ consists of a set $\mathrm{ob}(\mathcal{C})$ of objects and, for every pair of objects x and y, an object $\mathcal{C}(x, y)$ of $\mathcal{V}$ called a $\mathcal{V}$*-hom object* satisfying the following: for all objects $x, y, z \in \mathrm{ob}(\mathcal{C})$,

$$
\begin{aligned}
1 &\leq \mathcal{C}(x, x) \\
\mathcal{C}(y, z) \otimes \mathcal{C}(x, y) &\leq \mathcal{C}(x, z)
\end{aligned}
$$

Example 1. The unit interval $([0, 1], \leq, \cdot, 1)$ is a commutative monoidal preorder with the usual ordering $\leq$. Multiplication of real numbers is the monoidal product, which we will denote by juxtaposition, $ab := a \cdot b$ for $a, b \in [0, 1]$, and the monoidal unit is 1.

Example 2. The extended non-negative reals $([0, \infty], \geq, +, 0)$ form a commutative monoidal preorder where the preorder is the opposite of the usual ordering on the reals. The monoidal product is addition with $a + \infty := \infty$ and $\infty + a := \infty$ for all $a \in [0, \infty]$, and the monoidal unit is 0.

Sections **??** and **??** below give examples of categories enriched over both $[0, 1]$ and $[0, \infty]$ defined from the probabilities generated by a large language model. As we will see in the setup introduced in Subsections **??** and **??**, objects of both categories will consist of strings of characters from some finite token set.

2.4. **Every LLM defines a** $[0, 1]$**-category.** Equipped with π, we can now define a category $\mathcal{L}$ enriched over the unit interval. The objects of $\mathcal{L}$ coincide with the objects of L, and for a pair of objects x and y we define their $[0, 1]$-hom object to be $\mathcal{L}(x, y) := \pi(y|x)$.

Let us verify that $\mathcal{L}$ satisfies both the identity and compositionality requirements for a category enriched over $[0, 1]$ listed in Definition **??**. Suppose first that x, y, z are three strings satisfying $x \to y \to z$. If $x \neq y$ and $y \neq z$, then x and y are necessarily unfinished texts, and we may write

$$
\begin{aligned}
x &= \perp a_1 \cdots a_t \\
y &= \perp a_1 \cdots a_t \cdots a_{t+k} \\
z &= \perp a_1 \cdots a_t \cdots a_{t+k} \cdots a_{t+k+k'}.
\end{aligned}
$$

for some $(a_i)_{i=1}^{t+k+k'-1} \subset A$ and $a_{t+k+k'} \in A \cup \{\dagger\}$. It then follows from (**??**) that

$$\begin{aligned}
\pi(y|x)\pi(z|y) &= \prod_{i=1}^{k} p(a_{t+i}|y_{<t+i}) \prod_{j=1}^{k'} p(a_{t+k+j}|z_{<t+k+j}) \\
&= \prod_{i=1}^{k+k'} p(a_{t+i}|z_{<t+i}) \\
&= \pi(z|x).
\end{aligned} \tag{7}$$

If instead $x = y$, then $\pi(y|x) = 1$ and obviously $\pi(z|y) = \pi(z|x)$. The case $z = y$ is treated similarly. More generally, if x, y and z are arbitrary objects of L, one has that $\pi(y|x)\pi(z|y) \leq \pi(z|x)$ as it may be the case that $\pi(y|x), \pi(z|x) \neq 0$ but $\pi(z|y) = 0$, for example if $x = \perp$`green` and $y = \perp$`green lantern` and $z = \perp$`green salad`.

A more general version of this $[0,1]$-category was originally defined in [**?**, Definition 4], where $\mathcal{L}(x, y)$ was taken to be nonzero whenever x was an *arbitrary* substring of y (not strictly a prefix). However, the values $\pi(y|x)$ were not constructed explicitly from an LLM, whereas Definition **??** above yields a $[0,1]$-category of expressions in a language where the values $\pi(y|x)$ are described concretely from the probabilities generated by an LLM. A similar definition can be found in [**?**], although neither their description nor the one in [**?**] considers the special characters $\perp$ and $\dagger$ or a model's cutoff value. The incorporation of these aspects led us to the novel Proposition **??**, which *justifies referring to* π *as a probability.*

Notice we only consider extensions of expressions *on the right,* having in mind the most standard LLMs, although one could also consider bidirectional extensions. That said, the tree-like structure implied by the restriction to right extensions plays a key role in the computation of magnitude presented below.

Finally, let us also remark that, after defining the $[0,1]$-category $\mathcal{L}$ of strings, the authors of [**?**] further consider enriched copresheaves on $\mathcal{L}$, which were shown to contain semantic information. Our present goal, however, is to turn our attention away from $\mathcal{L}$ to a more geometric version of it.

2.5. **Every LLM defines a $[0,\infty]$-category.** As discussed in [**?**, Section 5], the function $-\ln\colon [0,1] \to [0,\infty]$ is an isomorphism of categories, which provides a passage from probabilities to a more geometric setting. That is, instead of considering an enrichment from probabilities involved in generating a string y from a prefix x, one could instead think of the "distance" traveled from x to y by defining

$$d(x, y) := -\ln \pi(y|x).$$

It is straightforward to check the triangle inequality is satisfied $d(x, y) + d(y, z) \geq d(x, z)$ and further that $d(x, x) = 0$ for all strings x, y, z. (This follows from Equation (**??**) and the fact that $\pi(x|x) = 1$ for all x.) From this perspective, a text that is highly likely to extend a prompt x is close to x, and a text that is not an extension of x is infinitely far way. In this way, we obtain a category $\mathcal{M}$ enriched over $[0,\infty]$, also known as a *generalized metric space* in the sense of Lawvere [**?**], by defining the $[0,\infty]$-hom object between a pair of strings x, y to be the distance $\mathcal{M}(x, y) := d(x, y)$. And now that we have a generalized metric space, we may inquire after its magnitude.

The theory behind the magnitude of (generalized) metric spaces is well known [?] and is a special case of the magnitude of enriched categories [?, ?]. In the next section, we provide a few preliminaries on magnitude and then compute the magnitude of $\mathcal{M}$. Importantly, the theory requires working with a *finite* enriched category, which is an additional reason to consider the category $\mathsf{L}^{\leq N}$ with finite cutoff N.

3. Magnitude

3.1. **Definition.** Magnitude is a numerical invariant of a category enriched over a monoidal category $(\mathcal{V}, \otimes, 1)$ [?, ?, ?]. Briefly, one starts with a semi-ring R and a multiplicative function $\|-\|\colon \mathrm{ob}(\mathcal{V}) \to R$ called *size* that is invariant under isomorphisms. So, $\|v\| = \|w\|$ whenever $v \cong w$ in $\mathcal{V}$, and $\|1\| = 1$ and $\|v \otimes w\| = \|v\|\|w\|$ for all $v, w \in \mathrm{ob}(\mathcal{V})$. Then, given a $\mathcal{V}$-category $\mathcal{C}$ with finitely many objects, one introduces the *zeta function* $\zeta_{\mathcal{C}}\colon \mathrm{ob}(\mathcal{C}) \times \mathrm{ob}(\mathcal{C}) \to R$, defined by $\zeta_{\mathcal{C}}(x, y) := \|\mathcal{C}(x, y)\|$. The function $\zeta_{\mathcal{C}}$ can be regarded as a square matrix with index set $\mathrm{ob}(\mathcal{C})$ [?]; when it is invertible, $\mathcal{C}$ is said to have *Möbius inversion* and its *Möbius function* is $\mu_{\mathcal{C}} := \zeta_{\mathcal{C}}^{-1}$ (the entries of this matrix are called *Möbius coefficients*). When $\mathcal{C}$ has Möbius inversion, the *magnitude* $\mathrm{Mag}(\mathcal{C})$ of $\mathcal{C}$ can be defined as

$$\mathrm{Mag}(\mathcal{C}) = \sum_{(x,y) \in \mathrm{ob}(\mathcal{C}) \times \mathrm{ob}(\mathcal{C})} \zeta_{\mathcal{C}}^{-1}(x, y). \tag{8}$$

For $\mathcal{V} = [0, \infty]$, every size function is of the form $\|x\|_t = e^{-tx}$ for some $t \in \mathbb{R} \cup \{\infty\}$. It is customary to choose $\|-\|_1$ and then consider the *magnitude function* $f(t) := \mathrm{Mag}(t\mathcal{C})$, defined for $t \in (0, \infty)$ [?, Section 2.2]. The symbol $t\mathcal{C}$ denotes the generalized metric space with the same objects as $\mathcal{C}$ but whose distances (that is, whose $[0, \infty]$-hom objects) are scaled by t.

3.2. **Magnitude of posets.** The magnitude of an enriched category stems from a classical story from posets, which we now briefly review from [?].

Given a poset P, define a *closed interval* to be $[s, t] = \{u \in P : s \leq u \leq t\}$ whenever $s \leq t$ in P. If every interval of P is finite, then P is said to be a *locally finite* poset.

Let $\Bbbk$ be a field, P a locally finite poset, and $\mathrm{int}(P)$ the set of all closed intervals of P. The *incidence algebra* $I(P, \Bbbk)$ of P over $\Bbbk$ is the $\Bbbk$-algebra of all functions $f\colon \mathrm{int}(P) \to \Bbbk$ with the usual structure of a vector space over $\Bbbk$, where multiplication is given by the *convolution product*,

$$(f * g)(s, u) := \sum_{s \leq t \leq u} f(s, t) g(t, u),$$

where we write $f(s, t)$ for $f([s, t])$. (Remark that the sum has finitely many terms.) The incidence algebra of P is an associative algebra with identity $\delta\colon \mathrm{int}(P) \to \Bbbk$ given by

$$\delta(s, u) = \begin{cases} 1 & \text{if } s = u \\ 0 & \text{if } s \neq u \end{cases}$$

since $f * \delta = \delta * f = f$ for all functions $f \in I(P, \Bbbk)$. The *zeta function* ζ_P is another notable function in the incidence algebra, defined to be constant at 1 on every interval,

$$\zeta_P(s, u) = 1, \quad \text{for all } s \leq u \text{ in } P.$$

In the context of language, the zeta function is quite simple.

Example 3. The zeta function of the poset L of strings is given by

$$\zeta_{\mathsf{L}}(x,y) = \begin{cases} 1 & \text{if } x \to y \\ 0 & \text{if } x \not\to y \end{cases}$$

for all x, y in L. This can also be seen by viewing L as a category enriched over truth values $\mathbf{2} := \{\texttt{true, false}\}$[3] and by considering the size function $\|-\|\colon \operatorname{ob}(\mathbf{2}) \to \mathbb{Z}$ given by $\|\texttt{true}\| = 1$ and $\|\texttt{false}\| = 0$ and then setting $\zeta_{\mathsf{L}}(x,y) = \|\mathsf{L}(x,y)\|$, which is 1 if x is a prefix of y and 0 otherwise.

For every locally finite poset P, the function ζ_P is an invertible element of the incidence algebra $\operatorname{int}(P)$ [**?**, Prop. 3.6.2]. Equivalently, the zeta function is invertible when regarded as a square matrix with index set P, cf. [**?**]; we have taken this matricial perspective in our presentation of the more general categorical definition in Subsection **??**. The inverse of ζ_P is called the *Möbius function* of P, denoted μ_P, and it can be computed recursively:

$$\mu_P(s,u) = \begin{cases} 1 & \text{if } s = u \\ -\displaystyle\sum_{s \le t < u} \mu_P(s,t) & \text{if } s < u \\ 0 & \text{otherwise.} \end{cases}$$

Example 4. The Möbius function μ_{L} on the poset L of strings is rather trivial. This is due to the tree-like structure of the full subcategory L_x of L introduced in Subsection **??**, whose objects are all strings y having a given string x as a prefix. Indeed, for any string $x \in \operatorname{ob}(\mathsf{L})$ and tokens $a_1, a_2, \ldots \in A \cup \{\dagger\}$, we have

$$\begin{aligned} \mu_{\mathsf{L}}(x,x) &= 1 \\ \mu_{\mathsf{L}}(x,xa_1) &= -\mu_{\mathsf{L}}(x,x) = -1 \\ \mu_{\mathsf{L}}(x,xa_1a_2) &= -\mu_{\mathsf{L}}(x,x) - \mu_{\mathsf{L}}(x,xa_1) = -1+1 = 0 \\ \mu_{\mathsf{L}}(x,xa_1a_2a_3) &= -\mu_{\mathsf{L}}(x,x) - \mu_{\mathsf{L}}(x,xa_1) - \mu_{\mathsf{L}}(x,xa_1a_2) = -1+1+0 = 0. \end{aligned}$$

Similarly, $\mu_{\mathsf{L}}(x, xa_1a_2\cdots a_j) = 0$ for $j > 1$.

This Möbius function may then be used to compute the magnitude of the finite category L of strings from a set A of tokens.

Example 5. Since the category L has an initial object, we know its magnitude is 1, cf. [**?**, Example 2.3], but it is also instructive to prove this via an explicit computation of the Möbius function. To that end, let $\#A$ be the cardinality of the token set A and let $\#\operatorname{ob}(\mathsf{L})$ be the number of strings in the finite poset L considered

[3]Every poset P defines a category enriched over truth values. Its objects are the elements of P, and given $x, y \in P$, if $x \le y$, then their corresponding hom object is $\texttt{true}$, and if $x \not\le y$ then it is $\texttt{false}$. The identity and compositionality requirements in Definition **??** arise from the reflexivity and transitivity of the partial order.

as a category. The magnitude of L is given by

$$\begin{aligned}\mathrm{Mag}(\mathsf{L}) &= \sum_{x,y\in \mathrm{ob}(\mathsf{L})} \zeta_{\mathsf{L}}^{-1}(x,y)\\ &= \sum_{x,y\in \mathrm{ob}(\mathsf{L})} \mu_{\mathsf{L}}(x,y)\\ &= \sum_{x\in \mathrm{ob}(\mathsf{L})} \mu_{\mathsf{L}}(x,x) + \sum_{\substack{x=\perp a\in \mathrm{ob}(\mathsf{L})\\ a\in \bigcup_{i=0}^{N-2} A^i}} \sum_{a_1\in A\cup\{\dagger\}} \mu_{\mathsf{L}}(x,xa_1)\\ &= \#\,\mathrm{ob}(\mathsf{L}) - \#\left\{x\in \mathrm{ob}(\mathsf{L}) : x=\perp a \text{ with } a\in \bigcup_{i=0}^{N-2} A^i\right\}(\#A+1).\end{aligned}$$

Unwinding the last line, recall that L denotes $\mathsf{L}^{\leq N} = \bigsqcup_{i=0}^{N-1} \mathsf{L}^{(i)}$. We have $\mathsf{L}^{(0)} = \{\perp\}$ and, for each $i \leq 1$, $\mathsf{L}^{(i)}$ is comprised of strings of the form $\perp a$ with $a \in A^i$ or $\perp a'\dagger$ with $a' \in A^{i-1}$. It follows that $\#\mathsf{L}^{(i)} = \#A^i + \#A^{i-1}$ and so

$$\begin{aligned}\#\,\mathrm{ob}(\mathsf{L}) &= 1 + \sum_{i=1}^{N-1} \#\mathsf{L}^{(i)}\\ &= 1 + \sum_{i=1}^{N-1} \#A^i + \#A^{i-1} = \#A^{N-1} + 2(\#A^{N-2}) + \cdots + 2(\#A) + 2.\end{aligned}$$

Furthermore,

$$\begin{aligned}\#\left\{x\in \mathrm{ob}(\mathsf{L}) : x=\perp a \text{ with } a\in \bigcup_{i=0}^{N-2} A^i\right\}(\#A+1) &= \left(\sum_{i=0}^{N-2} \#A^i\right)(\#A+1)\\ &= \#A^{N-1} + 2(\#A^{N-2}) + \cdots + 2(\#A) + 1.\end{aligned}$$

Therefore, $\mathrm{Mag}(\mathsf{L}) = 1$.

3.3. **Preliminaries on digraphs.** With this background in hand, we are on the way to computing the magnitude of the generalized metric space $\mathcal{M}$ associated with a large language model. The computation will rely on a recent combinatorial interpretation of the magnitude of a metric space given by Vigneaux in [**?**], which makes use of a directed graph associated with the space. For convenience, several definitions relating to directed graphs are provided in this section, followed by the main result in Subsection **??**.

A *directed graph*, or simply *digraph*, consists of a finite set V of *vertices* and a set $E \subset V \times V$ of *edges*. There are functions $s, t\colon E \to V$ defined on edges $e = (v, w)$ by $s(e) = v$ and $t(e) = w$ called *source* and *target*, respectively. When $s(e) = t(e)$, the edge is a called *loop*. The *out-degree* of a vertex v is the number of edges having v as the source, and the *in-degree* of v is the number of edges having v as the target. Any subset E' of E determines a new graph, called a *spanning subdigraph* (V, E'). A *weighted digraph* consists of a digraph together with numbers $w(e)$ called *weights* associated with each edge e.[4] Given a digraph $D = (V, E)$ and vertices $v, w \in V$,

[4]As an example, every poset $(P, \leq)$ defines a digraph with vertices given by elements of P and where there is an edge (p, q) whenever $p \leq q$ in P. One can also view this as a weighted digraph with weights 0 or 1, indicating whether or not $p \leq q$.

a *walk* from v to w is a sequence of edges $e_1, \ldots, e_k \in E$ for some $k \in \mathbb{N}$ such that $s(e_1) = v$ and $t(e_k) = w$ and $t(e_i) = s(e_{i+1})$ for $i = 1, \ldots, k-1$. The walk is said to have *length* k and *visits* vertices $x_i = t(e_i) = s(e_{i+1})$ for $i = 1, \ldots, k-1$. The walk is called a *cycle* if $v = w$ and $v, x_1, \ldots, x_k$ are all distinct, and it is called a *path* if $v, x_1, \ldots, x_k, w$ are all distinct.

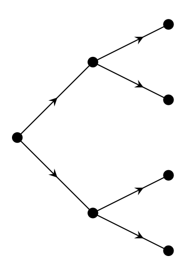
As we will soon see, the combinatorial interpretation of magnitude in [**?**] is based on a quotient of sums that involve particular subdigraphs—namely, linear subdigraphs and connections.

Definition 5. A *linear subdigraph* of a digraph D is a spanning subdigraph such that each vertex has in-degree 1 and out-degree 1. The set of all linear subdigraphs of D is denoted $L(D)$.

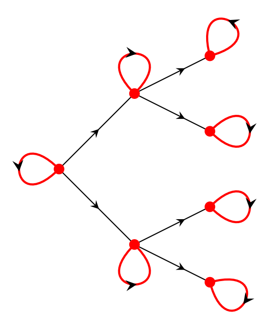
So a linear subdigraph amounts to a spanning collection of vertex-disjoint cycles. Now, given a linear subdigraph ℓ in a digraph (V, E), let $c(\ell)$ denote the number of cycles in ℓ and define its *signature* $\operatorname{sgn}(\ell)$ and *weight* $w(\ell)$, if the digraph is weighted, as follows:

$$\operatorname{sgn}(\ell) := (-1)^{\#V + c(\ell)} \qquad w(\ell) := \prod_{e \in \ell} w(e). \tag{9}$$

As an example, the digraph shown in Figure **??** does not have a linear subdigraph since the root vertex does not have in-degree 1, and the leaves do not have out-degree 1. If, however, the digraph were to have a loop at each vertex (for instance, if one were to think of the vertices as objects in a category) as shown in Figure **??**, then the collection of all loops comprises a linear subdigraph. It signature is $(-1)^{7+7} = 1$, and if a weighting had been assigned to each edge, then its weight would be a product over the seven loops.



(A) A digraph with no linear subdigraph.



(B) A linear subdigraph shown in red.

FIGURE 1. Visualizing a linear subdigraph.

Definition 6. A *connection* of a vertex v to a vertex w in a digraph D is a spanning subdigraph C in D with the following properties:

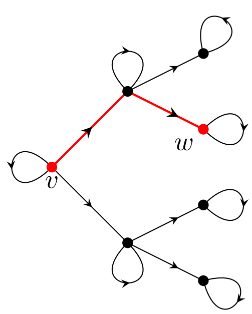
i. For all vertices $k \in V \setminus \{v, w\}$, the in-degree and out-degree of k is 1.
ii. There are no incoming edges at v and no outgoing edges at w.
iii. If $v \neq w$ then the out-degree of v is 1, and the in-degree of w is 1.

The set of all connections in D from v to w is denoted $C(D, v \to w)$.

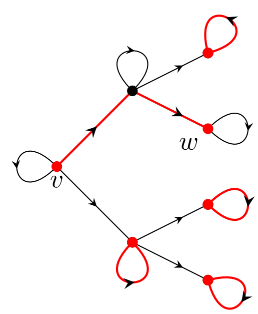
Given a connection C in a digraph (V, E), let $c(C)$ denote the number of cycles in C and define its *signature* $\operatorname{sgn}(C)$ and *weight* $w(C)$, if the digraph is weighted, as follows:

$$\operatorname{sgn}(C) := (-1)^{\#V + c(C) + 1} \qquad w(C) = \prod_{e \in C} w(e). \tag{10}$$

As an example, Figure **??** depicts a connection from v to w in the digraph displayed in Figure **??**. The signature of this connection is $(-1)^{7+4+1} = 1$, and if weights had been assigned to the edges of the digraph, then the weight of the connection would be a product over the four loops in the complement of the red path from v to w, multiplied by the weight of the path from v to w.



(A) A digraph and path from v to w.



(B) A connection (in red) from v to w.

FIGURE 2. Visualizing a connection.

Let S be a finite set and R a semiring. To any matrix $\xi : S \times S \to R$ we can associate a weighted directed graph $D(\xi)$ with vertex set S, whose directed edges are given by pairs $e = (x, y) \in S \times S$ such that $\xi(x, y) \neq 0$; the edge e has weight $w(e) = \xi(x, y)$. In particular, we can associate a weighted directed graph to the zeta function of an enriched $\mathcal{V}$-category in the sense of Section **??**, which is particularly useful in the case when $\mathcal{V} = [0, \infty]$. Indeed, connections and linear subdigraphs of the digraph associated with the generalized metric space $\mathcal{M}$ of strings will play an important role in the computation of magnitude in the next section.

3.4. **Main results.** We are now ready to compute the magnitude of the generalized metric space associated with an LLM. As before, let $\mathcal{M}$ be the $[0, \infty]$-category of strings introduced in Subsection **??**, whose objects are $\operatorname{ob}(\mathcal{M}) = \operatorname{ob}(\mathsf{L}^{\leq N})$ and where the $[0, \infty]$-hom objects are given by $\mathcal{M}(x, y) = d(x, y) = -\ln \pi(y|x)$. Define the *zeta matrix* $\zeta_{\mathcal{M}} \colon \operatorname{ob}(\mathcal{M}) \times \operatorname{ob}(\mathcal{M}) \to \mathbb{R}$ to be

$$\zeta_{\mathcal{M}}(x, y) := e^{-d(x,y)} = \pi(y|x),$$

and more generally for any real number $t > 0$ define

$$(\zeta_{\mathcal{M}})_t(x, y) := e^{-td(x,y)} = \pi(y|x)^t.$$

It is understood that $d(x, y) = \infty$ whenever $\pi(y|x) = 0$; for instance, a finished text is at infinite distance from any other string. For simplicity, we will write ζ_t instead of $(\zeta_{\mathcal{M}})_t$. As we will see in the proposition and corollaries below, ζ_t^{-1} exists and and its components can be computed explicitly.

Proposition 2. *For any $t > 0$, the matrix ζ_t is invertible. Moreover, for any x, y in* $\mathrm{ob}(\mathcal{M})$,

$$\zeta_t^{-1}(x,y) = \sum_{k\geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0\to y_1\to\cdots\to y_k=y}} \text{in } \mathsf{L} \quad (-1)^k \prod_{i=1}^{k} \pi(y_i|y_{i-1})^t$$

Proof. The main reference for this proof is [?], which relies on the graph-theoretic notions introduced above. Consider the weighted digraph $D(\zeta_t)$. By Proposition 3.11 in [?],

$$\zeta_t^{-1}(x,y) = \frac{1}{\det \zeta_t} \sum_{C\in C(D(\zeta_t),x\to y)} \mathrm{sgn}(C)w(C)$$

where the determinant of the zeta matrix is equal to

$$\det \zeta_t = \sum_{\ell\in L(D(\zeta_t))} \mathrm{sgn}(\ell)w(\ell).$$

Recall that $L(D(\zeta_t))$ is the set of all linear subdigraphs of $D(\zeta_t)$ and $C(D(\zeta_t), x \to y)$ is the set of all connections in $D(\zeta_t)$ from x to y (cf. Definitions **??** and **??**).

Since there is an edge (x, y) in the digraph associated to ζ_t if and only if $x \to y$ in the ordinary category L, one can restrict the sum comprising $\det \zeta_t$ to linear subdigraphs of L, which are spanning collections of vertex-disjoint cycles. Recall that a cycle consists of texts $x = y_0 \to y_1 \to \cdots \to y_k = x$ where the y_i are all distinct for $i = 0, \ldots, k-1$. Thus, the only linear subdigraph of L is given by loops $x \to x$ on each object $x \in \mathrm{ob}(\mathsf{L})$. By Equation (**??**), this linear subdigraph has weight 1 and signature equal to $(-1)^{\#\,\mathrm{ob}(\mathsf{L})+\#\,\mathrm{ob}(\mathsf{L})} = 1$.

In turn, the Möbius coefficients simplify to

$$\zeta_t^{-1}(x,y) = \sum_{C\in C(D(\zeta_t),x\to y)} \mathrm{sgn}(C)w(C).$$

Now observe that a connection C from x to y in $\mathcal{M}$ consists of a nondegenerate path $x = y_0 \to y_1 \to \cdots = y_k = y$ in L and a linear subdigraph ℓ_C of its complement, which, as noted above, is comprised of loops. By Equation (**??**), the weight of such a connection is[5]

$$\begin{aligned} w(C) &= \zeta_t(x,y_1)\zeta_t(y_1,y_2)\cdots\zeta_t(y_{k-1},y)w(\ell_C) \\ &= \pi(y_1|y_0)^t\pi(y_2|y_1)^t\cdots\pi(y_k|y_{k-1})^t\cdot 1 \end{aligned}$$

and its signature is $\mathrm{sgn}(C) = (-1)^{\#\,\mathrm{ob}(\mathcal{M})+c(C)+1}$ which is the same as $(-1)^{\#\,\mathrm{ob}(\mathcal{M})-c(C)+1}$ and, since the number of cycles $c(C)$ in the connection is equal to the number of objects in the complement of the path $x = y_0 \to y_1 \to \cdots = y_k = y$, this is equal to $(-1)^k$. □

The following corollary shows that the Möbius coefficients obtained from ζ_t^{-1} have a simplified expression in terms of π from Definition **??** and the Möbius function ζ_{L}^{-1}, whose zeta function ζ_{L} was described in Example **??**.

[5]If $y_0 \to \cdots \to y_k$ is not a path in L then it has weight 0 and doesn't contribute to the sum.

Corollary 1. *For any strings x, y in* $\mathrm{ob}(\mathcal{M})$*,*

$$\zeta_t^{-1}(x,y) = \pi(y|x)^t\ \zeta_{\mathsf{L}}^{-1}(x,y).$$

Proof. For any $k \geq 0$ and any nondegenerate path $x = y_0 \to y_1 \to \cdots \to y_k = y$ in L, Equation (**??**) implies

$$\begin{aligned}\prod_{i=1}^{k} \pi(y_i|y_{i-1})^t &= \pi(y_1|y_0)^t \pi(y_2|y_1)^t \pi(y_3|y_2)^t \cdots \pi(y_k|y_{k-1})^t \\ &= \pi(y_2|y_0)^t \pi(y_3|y_2)^t \cdots \pi(y_k|y_{k-1})^t \\ &= \quad \vdots \\ &= \pi(y_k|y_0)^t.\end{aligned}$$

Hence the Möbius coefficients $\zeta_t^{-1}(x,y)$ can be written as

$$\pi(y|x)^t \sum_{k\geq 0} (-1)^k \#\{\text{nondegenerate paths of length } k \text{ from } x \text{ to } y \text{ in } \mathsf{L}\}$$

which is equal to $\pi(y|x)^t \zeta_{\mathsf{L}}^{-1}(x,y)$ by Philip Hall's theorem [**?**], cf. [**?**, Proposition 3.8.5], [**?**, Corollary 1.5]. □

Now recall from Example **??** that $\zeta_{\mathsf{L}}^{-1}(x,y) = \mu_{\mathsf{L}}(x,y)$, which shows that ζ_t^{-1} can be simplified even further.

Corollary 2. *For any strings x, y in* $\mathrm{ob}(\mathsf{L})$*,*

$$\zeta_t^{-1}(x,y) = \begin{cases} -\pi(y|x)^t & \text{if } y \in \mathsf{L}_x^{(1)} \\ 1 & \text{if } y = x \\ 0 & \text{otherwise.} \end{cases}$$

Finally, then, we may use these results to write down the *magnitude function* of the generalized metric space $\mathcal{M}$. By Equation (**??**) it is the function $f : (0,\infty) \to \mathbb{R}$ given by $f(t) = \mathrm{Mag}(t\mathcal{M})$, where[6]

$$\mathrm{Mag}(t\mathcal{M}) := \sum_{x,y \in \mathrm{ob}(\mathcal{M})} \zeta_t^{-1}(x,y).$$

Now recall that for any real number t, the *Tsallis entropy* of a probability distribution $p = (p_1, \ldots, p_n)$ is equal to

$$H_t(p) = \frac{1}{t-1}\left(1 - \sum_{i=1}^{n} p_i^t\right).$$

The following proposition expresses the magnitude function of $\mathcal{M}$ in terms of the Tsallis entropies of the probability distributions $(p_x \colon A \cup \{\dagger\} \to [0,1])_{x \in \mathrm{ob}(\mathsf{L})}$ generated by a large language model and the cardinality of the terminal states $T(\perp)$ of the beginning-of-sentence symbol.

[6] By Corollary **??**, it follows that $\mathrm{Mag}(t\mathcal{M}) = \sum_{x,y\in\mathrm{ob}(\mathcal{M})} \pi(y|x)^t \zeta_{\mathsf{L}}^{-1}(x,y)$, which is a weighted version of the magnitude of the poset L in Example **??**.

Proposition 3. *The magnitude function of the generalized metric space $\mathcal{M}$ associated with an LLM is equal to*

$$\mathrm{Mag}(t\mathcal{M}) = (t-1) \sum_{x \in \mathrm{ob}(\mathcal{M}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \qquad t > 0.$$

Proof. By Corollary **??**,

$$\mathrm{Mag}(t\mathcal{M}) = \sum_{x,y \in \mathrm{ob}(\mathcal{M})} \zeta_t^{-1}(x,y)$$

$$= \sum_{x \in \mathrm{ob}(\mathcal{M})} \zeta_t^{-1}(x,x) - \sum_{x \in \mathrm{ob}(\mathcal{M}) \setminus T(\perp)} \sum_{y \in \mathsf{L}_x^{(1)}} \pi(y|x)^t \tag{11}$$

$$= \sum_{x \in \mathrm{ob}(\mathcal{M}) \setminus T(\perp)} \left(1 - \sum_{a \in A \cup \{\dagger\}} p_x(a)^t \right) + \#(T(\perp)) \tag{12}$$

Observe that the first sum in Equation (**??**) is over all objects in $\mathcal{M}$, including those strings x that are not the prefix of any other string $y \neq x$, such as all finished texts. The double sum in Equation (**??**) accounts for unfinished strings and their extensions by a single token, which is why the indexing set is $\mathrm{ob}(\mathcal{M}) \setminus T(\perp)$. Equation (**??**) follows from the decomposition $\mathrm{ob}(\mathcal{M}) = T(\perp) \sqcup (\mathrm{ob}(\mathcal{M}) \setminus T(\perp))$ and the fact that $\zeta_t^{-1}(x,x) = 1$ for all $x \in \mathrm{ob}(\mathcal{M})$.

When $t = 1$, Equation (**??**) is simply $\#(T(\perp))$, since p_x is a probability mass function on $A \cup \{\dagger\}$. When $t \neq 1$, we can multiply the sum by $(t-1)/(t-1)$ and rearrange factors to conclude. □

Remark 1. It is easy to show (e.g. via L'Hôpital's rule) that if $p : S \to [0,1]$ is a probability mass function on a finite set S,

$$\lim_{t \to 1} H_t(p) = \lim_{t \to 1} \frac{1}{t-1} \left(1 - \sum_{s \in S} p(s)^t \right) = -\sum_{s \in S} p(s) \ln p(s) =: H(p).$$

The quantity $H(p)$ is the *Shannon entropy* of p (in nats). This entails that Shannon entropy emerges when computing the derivative of the magnitude function $f(t) = \mathrm{Mag}(t\mathcal{M})$ at $t = 1$:

$$f'(1) = \sum_{x \in \mathrm{ob}(\mathcal{M}) \setminus T(\perp)} H(p_x). \tag{13}$$

Alternatively, we can deduce from Equation (**??**) that

$$f(t) = \#(\mathrm{ob}(\mathcal{M})) - \sum_{x \in \mathrm{ob}(\mathcal{M}) \setminus T(\perp)} Z_x(t),$$

where

$$Z_x(t) = \sum_{a \in A \cup \{\dagger\}} p_x(a)^t = \sum_{a \in A \cup \{\dagger\}} e^{-t(-\ln p_x(a))}$$

is the *partition function* of a system with microstates $A \cup \{\dagger\}$ and internal energy $E_x(a) = -\ln p_x(a) = d(x, xa)$ at inverse temperature $t > 0$. One can verify (see, for instance, [**?**]) that

$$H(p_x) = -\left.\frac{\mathrm{d}}{\mathrm{d}t} Z_x(t)\right|_{t=1},$$

from which Equation (**??**) also follows.

Since internal energy does not have to be normalized, we could also write $f(t) = \#(\mathrm{ob}(\mathcal{M})) - \tilde{Z}(t)$, where $\tilde{Z}$ is the partition function of a system with microstates $(x,a) \in S := (\mathrm{ob}(\mathcal{M}) \setminus T(\perp)) \times (A \cup \{\dagger\})$ and internal energy $E(x,a) = d(x,a) = -\ln p_x(a)$. The set S parameterizes indecomposable non-identity arrows in L. For each $t > 0$, there is a corresponding probability mass function (Gibbs state) ρ on S, given by $\rho(s) = e^{-tE(s)}/\tilde{Z}(t)$. Then

$$\mathbb{E}_\rho(E) = \sum_{s\in S} E(s)\frac{e^{-tE(s)}}{\tilde{Z}(t)} = -\frac{\mathrm{d}}{\mathrm{d}t}\ln \tilde{Z}(t) = \frac{f'(t)}{\tilde{Z}(t)},$$

which gives yet another interpretation of the derivative of the magnitude function.

Remark 2. If desired, one can also make sense of the magnitude function of an enriched category associated with a particular string. That is, suppose $x \in \mathrm{ob}(\mathcal{M})$ and let $\mathcal{M}_x$ denote the $[0,\infty]$-category whose objects are strings y containing x as a prefix. Define the hom-object between any $y, z \in \mathrm{ob}(\mathcal{M}_x)$ to be $\mathcal{M}_x(y,z) := \mathcal{M}(y,z) = -\ln \pi(z|y)$. Compositionality and the identity requirement hold since they hold in $\mathcal{M}$. So $\mathcal{M}_x$ is indeed a $[0,\infty]$-category and its magnitude function is given by

$$\zeta_{\mathcal{M}_x} = \zeta_{\mathcal{M}}\big|_{\mathrm{ob}(\mathcal{M}_x)\times\mathrm{ob}(\mathcal{M}_x)}$$

and because of Proposition **??**, its inverse is equal to

$$\zeta^{-1}_{\mathcal{M}_x} = \zeta^{-1}_{\mathcal{M}}\big|_{\mathrm{ob}(\mathcal{M}_x)\times\mathrm{ob}(\mathcal{M}_x)},$$

which implies

$$\begin{aligned}\mathrm{Mag}(t\mathcal{M}_x) &= \sum_{y,z\in\mathrm{ob}(\mathcal{M}_x)} \zeta^{-1}_{\mathcal{M}_x}(y,z)\\ &= \sum_{y\in\mathrm{ob}(\mathcal{M}_x)} \zeta^{-1}_{\mathcal{M}_x}(y,y) - \sum_{y\in\mathrm{ob}(\mathcal{M}_x)\setminus T(x)}\ \sum_{y\in\mathsf{L}^{(1)}_y} \pi(z|y)^t\\ &= (t-1)\sum_{y\in\mathrm{ob}(\mathcal{M}_x)\setminus T(x)} H_t(p_y) + \#(T(x)).\end{aligned}$$

3.5. **Magnitude homology.** The magnitude homology of metric spaces and enriched categories has been studied extensively by Leinster and Shulman in [**?**]. The main theorem of their work, namely [**?**, Theorem 7.14], expresses the magnitude function of a metric space (M,d) as a weighted sum of Euler characteristics of certain chain complexes. More precisely, for each $\ell \in [0,\infty)$, there is a chain complex comprised of free abelian groups $(MC_{k,\ell}(M))_{k\in\mathbb{N}}$ such that $MC_{k,\ell}(M) := \mathbb{Z}[G_{k,\ell}]$ where

$$G_{k,\ell} = \left\{(y_0,\dots,y_k) \in M^{k+1} \mid \sum_{i=0}^{k-1} d(y_i,y_{i+1}) = \ell \text{ and for all } i,\ y_i \neq y_{i+1}\right\}.$$

The differential $\partial_k : MC_{k,\ell}(M) \to MC_{k-1,\ell}(M)$ is defined as an alternating sum

$$\partial_k = \sum_{i=0}^{k}(-1)^i\partial_k^i$$

where for each $0 < i < k$,

$$\partial_k^i(y_0, \dots, y_k) = \begin{cases} (y_0, \dots, y_{i-1}, y_{i+1}, \dots, y_k) & \text{if } d(y_{i-1}, y_i) + d(y_i, y_{i+1}) = d(y_{i-1}, y_{i+1}) \\ 0 & \text{otherwise} \end{cases}$$

and $\partial_k^0 = \partial_k^k = 0$. The resulting $[0, \infty)$-graded chain complex $(MC_{\bullet,\ell}(M), \partial_\bullet)$ is called the *magnitude complex* of M [**?**, Definition 3.3] and its homology $H_{\bullet,\bullet}(M)$ is called *magnitude homology* [**?**, Definition 3.4]. The same definitions hold for any generalized metric space, such as $\mathcal{M}$.

The following corollary expresses the magnitude function of $\mathcal{M}$ as a weighted sum of Euler characteristics. It is a direct result of Proposition **??** and also mirrors [**?**, Theorem 7.14], although the situation is considerably simpler here because each complex $(MC_{\bullet,\ell})_\ell$ is bounded and moreover only finitely many ℓ can arise as total lengths, which implies we only have to deal with finite sums instead of infinite series.

Proposition 4. *For any $t > 0$, set $q = e^{-t}$. Then*

$$\mathrm{Mag}(t\mathcal{M}) = \sum_\ell q^\ell \sum_{k \geq 0} (-1)^k \operatorname{rank}(H_{k,\ell}(\mathcal{M}))$$

where the first sum is over those finitely many values of $\ell \in [0, \infty)$ for which $\bigcup_{k \geq 0} MC_{k,\ell}(\mathcal{M}) \neq \emptyset$.

Proof. By combining the definition of the magnitude function and its expression in Proposition **??**, we have

$$\begin{aligned} \mathrm{Mag}(t\mathcal{M}) &= \sum_{x,y \in \mathrm{ob}(\mathcal{M})} \zeta_t^{-1}(x, y) \\ &= \sum_{x,y \in \mathrm{ob}(\mathcal{M})} \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x = y_0 \to y_1 \to \dots \to y_k = y}} {}_{\text{in } \mathsf{L}} (-1)^k \prod_{i=1}^k \pi(y_i | y_{i-1})^t. \end{aligned}$$

Observe that the sum over nondegenerate paths can be restricted to those paths $x = y_0 \to y_1 \to \dots \to y_k = y$ such that $\prod_{i=1}^k \pi(y_i|y_{i-1}) \neq 0$, which implies that $d(y_i, y_{i+1}) < \infty$ for any $i = 0, \dots, k-1$. Further, there is a bijective correspondence between nondegenerate paths $x = y_0 \to y_1 \to \dots \to y_k = y$ of total length $\ell := \sum_{i=0}^{k-1} d(y_i, y_{i+1}) < \infty$ and generators $(y_0, ..., y_k)$ in $G_{k,\ell}$. Now remark that $\prod_{i=1}^k \pi(y_i|y_{i-1})^t = \exp(-t\ell)$, and so it is natural to group all the paths that have the same total distance, resulting in the following:

$$\mathrm{Mag}(t\mathcal{M}) = \sum_{k \geq 0} \sum_{\ell \in [0,\infty)} \sum_{c \in G_{k,\ell}} (-1)^k e^{-t\ell}. \tag{14}$$

Since $\mathcal{M}$ is finite, only finitely many real numbers $\ell \in [0, \infty)$ can arise as total lengths of nondegenerate paths, which clarifies the meaning of the sum second sum. Moreover $\sum_{c \in G_{k,\ell}} (-1)^k e^{-t\ell} = (-1)^k q^\ell \# G_{k,\ell} = (-1)^k q^\ell \operatorname{rank}(MC_{k,\ell}(\mathcal{M}))$. The combination of this fact and Equation (**??**) yields the claim by following a classical argument present in the proof of [**?**, Theorem 6.17], for instance; it is important that each group $MC_{k,\ell}$ is finitely generated and that $(MC_{\bullet,\ell})_\ell$ is bounded. $\square$

Remark 3. Proposition **??** expresses the magnitude function as a weighted sum of Euler characteristics. Together with Proposition **??**, it establishes a connection between entropy and topological invariants, perhaps sitting alongside other recent results linking information theory and algebraic topology [**?**, **?**, **?**, **?**].

Finally, we make a few additional observations about the *magnitude homology* of $\mathcal{M}$, which is the homology of the magnitude complex defined above. Although Leinster and Shulman work primarily with metric spaces in [**?**], two of their results translate directly to our setting:

(1) The group $H_{0,0}(\mathcal{M})$ is the free abelian group on $\mathrm{ob}(\mathcal{M})$, and for any $\ell > 0$, $H_{0,\ell}(\mathcal{M}) = 0$, cf. [**?**, Theorem 4.1].
(2) The group $H_{1,\ell}(\mathcal{M})$ is the free abelian group on the set of ordered pairs (x, y) such that $x \neq y$ and $d(x, y) = \ell$ and there does not exist any point strictly between x and y in L, cf. [**?**, Theorem 4.3].

Given these facts, one rewrite Equation (**??**) as

$$\mathrm{Mag}(t\mathcal{M}) = \operatorname{rank} H_{0,0}(\mathcal{M}) - \sum_{\ell \geq 0} q^{\ell} \operatorname{rank} H_{1,\ell}(\mathcal{M}). \tag{15}$$

We conjecture that higher homology groups vanish. If this is the case, Equation (**??**) would also follow Proposition **??**.

4. Final remarks

We conclude with a few remarks surrounding the results of this article. To start, one might notice that the zeta function associated with the generalized metric space $\mathcal{M}$ relates to perplexity, which is frequently used in evaluating autoregressive language models. To elaborate, the *perplexity* of a tokenized sequence $y = a_0 a_1 \cdots a_n$ is

$$PPL(x) = \exp\left\{ -\frac{1}{n} \sum_{i=1}^{n} \ln p(a_i | y_{<i}) \right\}$$

where $p(-|y_{<i})$ is the next-token probability distribution generated by the model when prompted with $y_{<i}$ [**?**], cf Subsection **??**. Perplexity is thus equal to the reciprocal of the zeta function $\zeta_t(a_0, y)$ when $t = 1/n$, and a_0 is the first token (usually $\perp$), and y is the full string:

$$PPL(y) = 1/\zeta_t(a_0, y) = 1/e^{t \ln \pi(y|a_0)},$$

or said differently, $\zeta_t(a_0, y) = 1/PPL(y)$. So perhaps for intuition, one might wish to think of the zeta function $\zeta_t(x, y)$ for arbitrary x, y as a generalization of the (reciprocal) of perplexity.

On a different note, as remarked in the introduction, one advantage to defining an enriched category of strings from a language is that one can pass to enriched *co-presheaves* on those strings, and the latter functor category contains rich structure. Concretely, there is a $[0, \infty]$-category $\widehat{\mathcal{M}} := [0, \infty]^{\mathcal{M}}$ whose objects are $[0, \infty]$-copresheaves, that is, functions $f\colon \mathcal{M} \to [0, \infty]$ satisfying $\max\{f(y) - f(x), 0\} \leq \mathcal{M}(x, y)$ for all strings x and y. The relevance is that while $\mathcal{M}$ does not, a priori, have any sort of algebraic structure, the functor category $\widehat{\mathcal{M}}$ does, in the sense that one can compute weighted limits and colimits between copresheaves that are reminiscent of logical operations on meaning representations of texts [**?**, Section 5].

It may be interesting to compute the magnitude of an appropriate finite version of $\widehat{\mathcal{M}}$, cf. [**?**, Example 2.5].

Lastly, let us draw a brief connection to the notion of diversity. *Diversity* depends on a finite category A, a probability mass function $p\colon \mathrm{ob}(\mathsf{A}) \to [0,1]$, and a similarity matrix $\theta\colon \mathrm{ob}(\mathsf{A}) \times \mathrm{ob}(\mathsf{A}) \to [0,\infty)$ such that

$$\theta(a,b) = 0 \quad \text{if} \quad \mathsf{A}(a,b) = \varnothing.$$

In the case of a generalized metric space, $\theta = \zeta_t$ and diversity is given by[7]

$$\begin{aligned}\mathscr{H}(\mathcal{M}^{op}, p, \zeta) &= -\sum_{y\in \mathrm{ob}(\mathcal{M})} p(y)\log\left(\sum_{x\in\mathrm{ob}(\mathcal{M})} \zeta_t^T(y,x)p(x)\right)\\ &= -\sum_{y\in \mathrm{ob}(\mathcal{M})} p(y)\log\left(\sum_{x\in\mathrm{ob}(\mathcal{M})} \pi(y|x)^t p(x)\right).\end{aligned}$$

In [**?**] the authors make a case for seeing this function as a probabilistic extension of log(magnitude), just as entropy is a probabilistic extension of log(cardinality) already in the work of Boltzmann and Gibbs. We leave a detailed study of diversity for future work, but remark here that if we take $p = \pi(-|x)|_{T(x)}$, then we recover the Shannon entropy of $\pi(-|x)|_{T(x)}$, which unlike our expression for the magnitude does not treat all states equally and gives more weight to those that are highly probable.

Acknowledgments

The authors thank Matilde Marcolli and Yiannis Vlassopoulos for engaging in numerous insightful discussions on category theory, language models, and linguistics that have influenced this article.

SandboxAQ, Palo Alto, CA 94301, USA

Department of Mathematics, The Master's University, Santa Clarita, CA 91321, USA
Email address: `tai.danae@math3ma.com`

Department of Mathematics, California Institute of Technology, Pasadena, CA 91125, USA
Email address: `vigneaux@caltech.edu`

[7]The magnitude of $\mathcal{M}$ is equal to the magnitude of the opposite $[0,\infty]$-category $\mathcal{M}^{op}$, where $[0,\infty]$-hom objects are given by $\mathcal{M}^{op}(x,y) = \mathcal{M}(y,x) = \pi(x|y)$. This follows since paths in $\mathcal{M}$ are in one-to-one correspondence with paths in $\mathcal{M}^{op}$, and the only paths with nontrivial weights come from paths in the poset L (respectively L^{op}). That is, $y_0 \to \cdots \to y_k$ is a path in L if and only if $y_k \to \cdots \to y_0$ is a path in L^{op} with weights $\mathcal{M}^{op}(y_k, y_{k-1}), \ldots, \mathcal{M}^{op}(y_1, y_0)$, which are equal to $\mathcal{M}^{op}(y_0, y_1), \ldots, \mathcal{M}^{op}(y_{k-1}, y_k)$, that is, $\pi(y_1|y_0), \ldots, \pi(y_k|y_{k-1})$.